Sandboxing and resource limits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Sandboxing and resource limits before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Untrusted input boundaries

Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

State machines for bounded agents

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Sandboxing and **quota**s constrain what a **process** may access or consume. They reduce possible operations but do not verify every result.

The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.

Practise with fading help

> Complete the supported Sandboxing and resource limits trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Sandbox**ing and resource limits. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Sandboxing and resource limits explanation against the supplied facts before discussing it.

Use the worked case for **Sandbox**ing and resource limits. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Queue currently has 9 tasks; job proposes 2 more under a maximum 10 policy. Decide.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

Explain the mechanism. Apply the supplied Sandboxing and resource limits facts and keep

> your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Sandboxing and resource limits facts and keep your reasoning.

A job requests a public webpage and a write outside its **output** directory. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Sandboxing and resource limits facts and keep your reasoning.

A report omits this qualification: Resource limits constrain execution but cannot prove generated claims or algorithms correct. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Sandboxing and resource limits facts and keep your reasoning.

Queue currently has 9 tasks; job proposes 2 more under a maximum 10 policy. Decide.

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Sandboxing and resource limits facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Sandboxing and resource limits to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **sandbox**ing and resource limits **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **sandbox**ing and resource limits.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

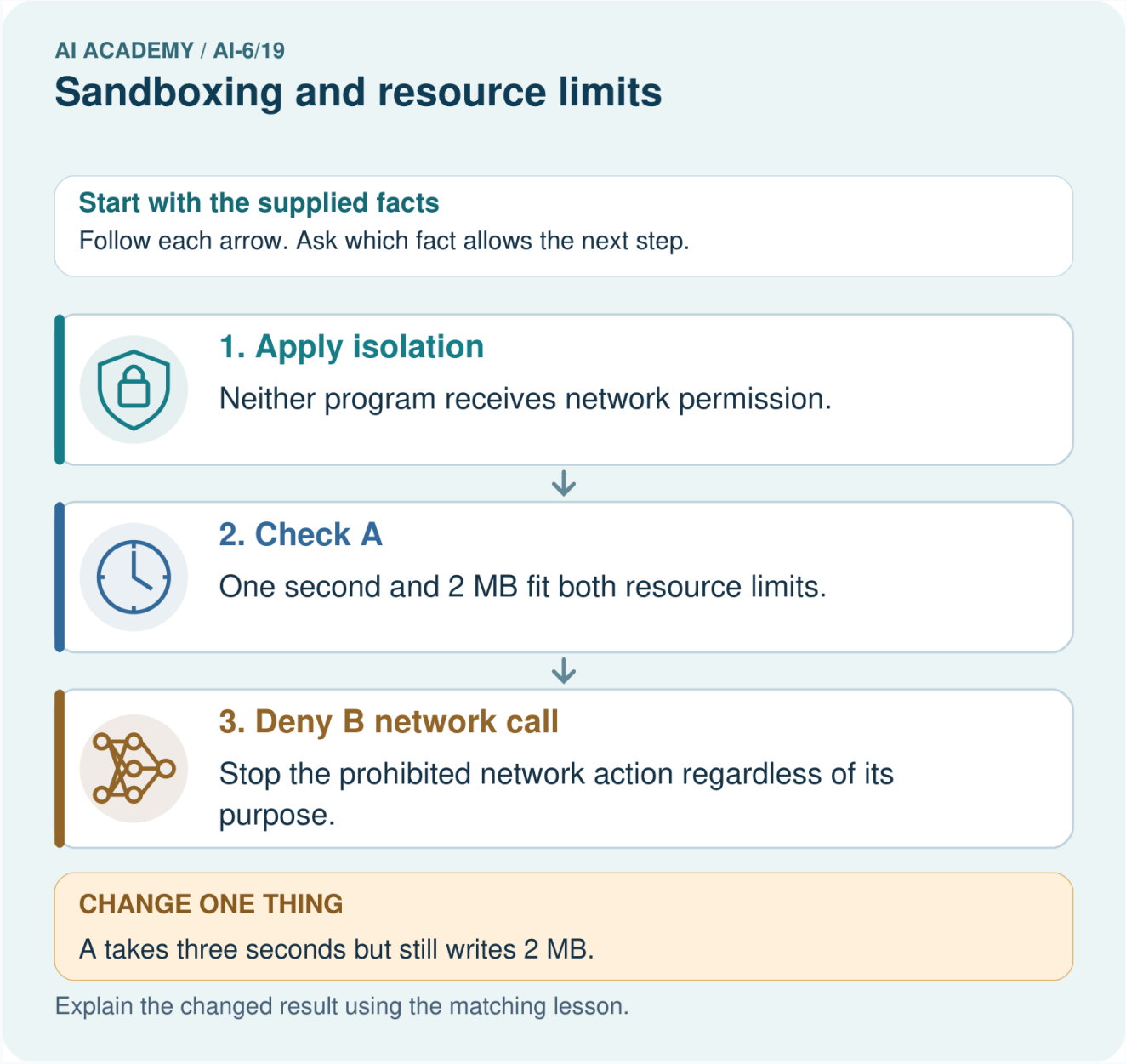
Pictorial case practice: Sandboxing and resource limits

> Bound a student program. A toy sandbox forbids network access, allows at most two seconds

> and limits output to 20 MB. Explain the labelled step before checking the changed condition.

Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB.

A uses one second and writes 2 MB. B tries to contact a website. These are declared trial records.



Follow the labelled picture with the matching explanation. Open the vector diagram

Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its **output** remains only 2 MB?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

Keep this week’s input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Stop the prohibited network action regardless of its purpose.

1. Deny B network call

2. Apply isolation

3. Check A

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Deny B network call Stop the prohibited network action regardless of its purpose. **Sandbox** controls bound reach and resource use.

Passing an **output** limit cannot cancel a network or time restriction.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A takes three seconds but still writes 2 MB. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Stop A for exceeding the time allowance.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop A for exceeding the time

allowance. The independent two-second limit remains binding.

Repair a missing building block

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

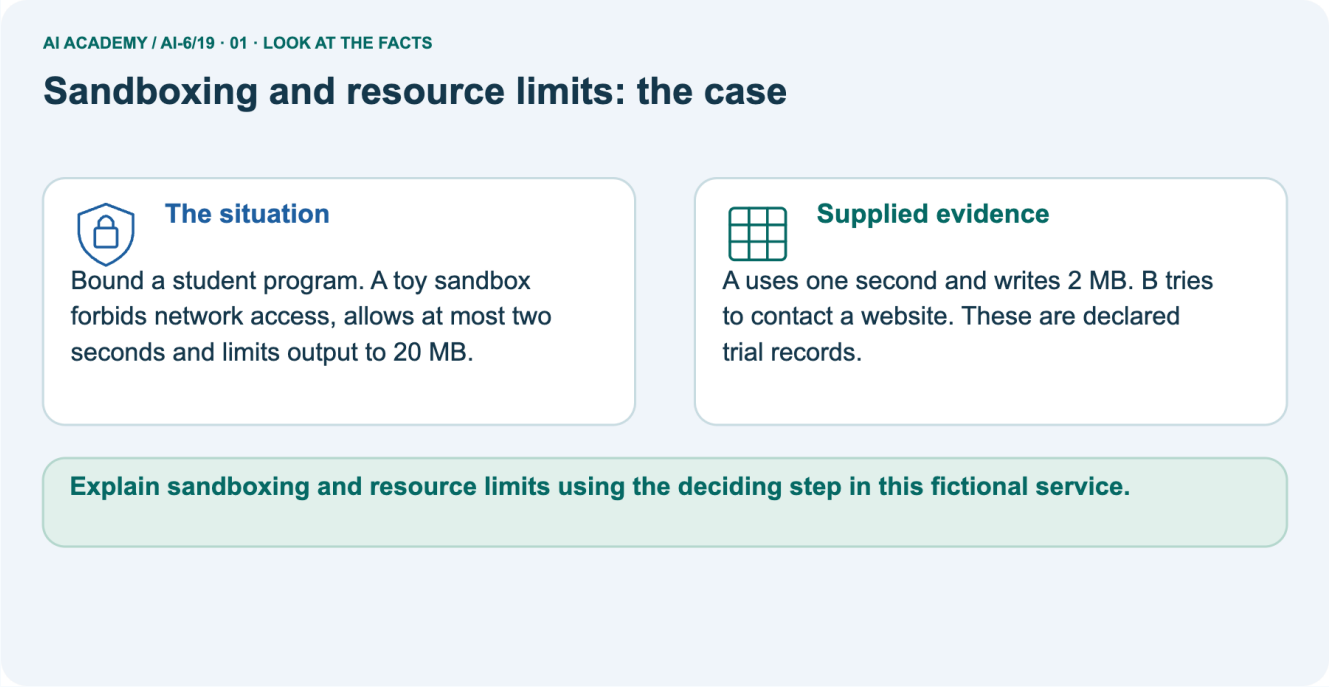
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Sandboxing and resource limits picture from its labels, then record a relevant limit.

01. Sandboxing and resource limits: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

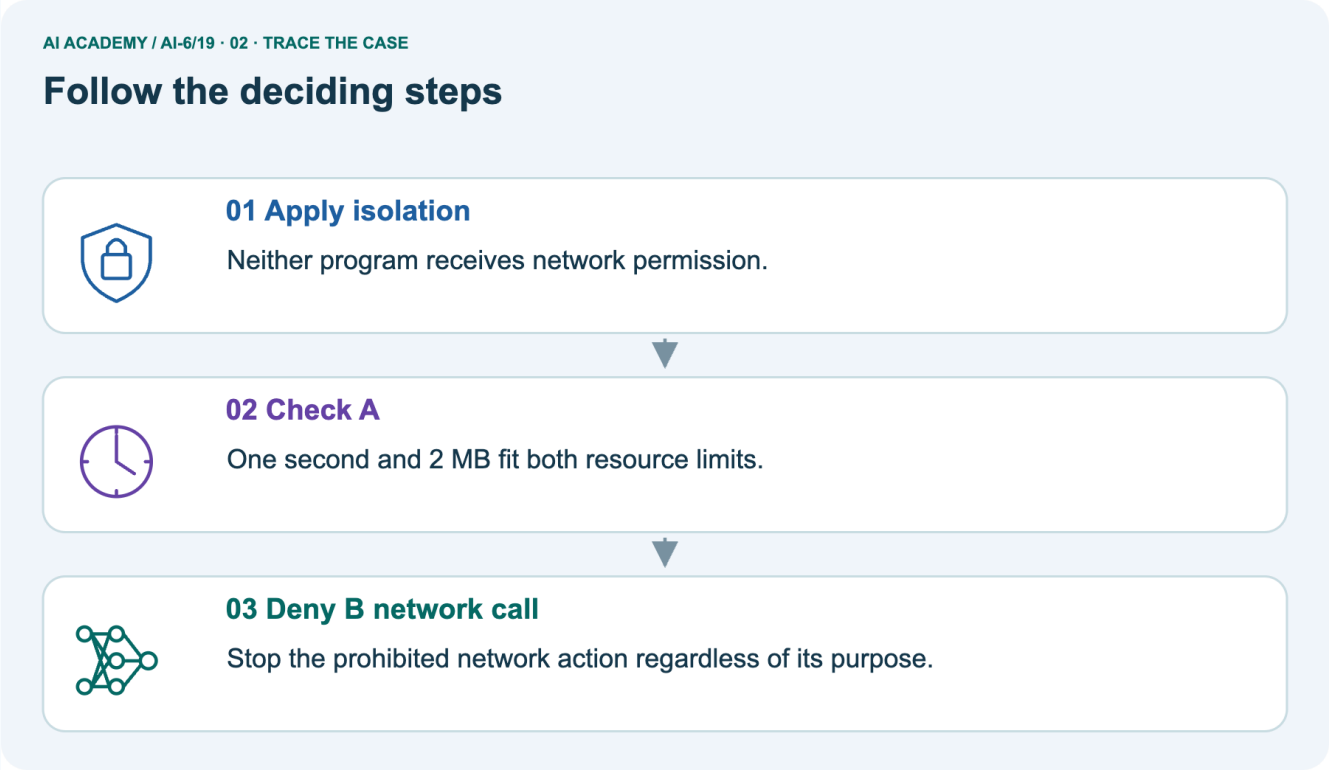
Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB. Supplied facts: A uses one second and writes 2 MB. B tries to contact a website.

These are declared trial records.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



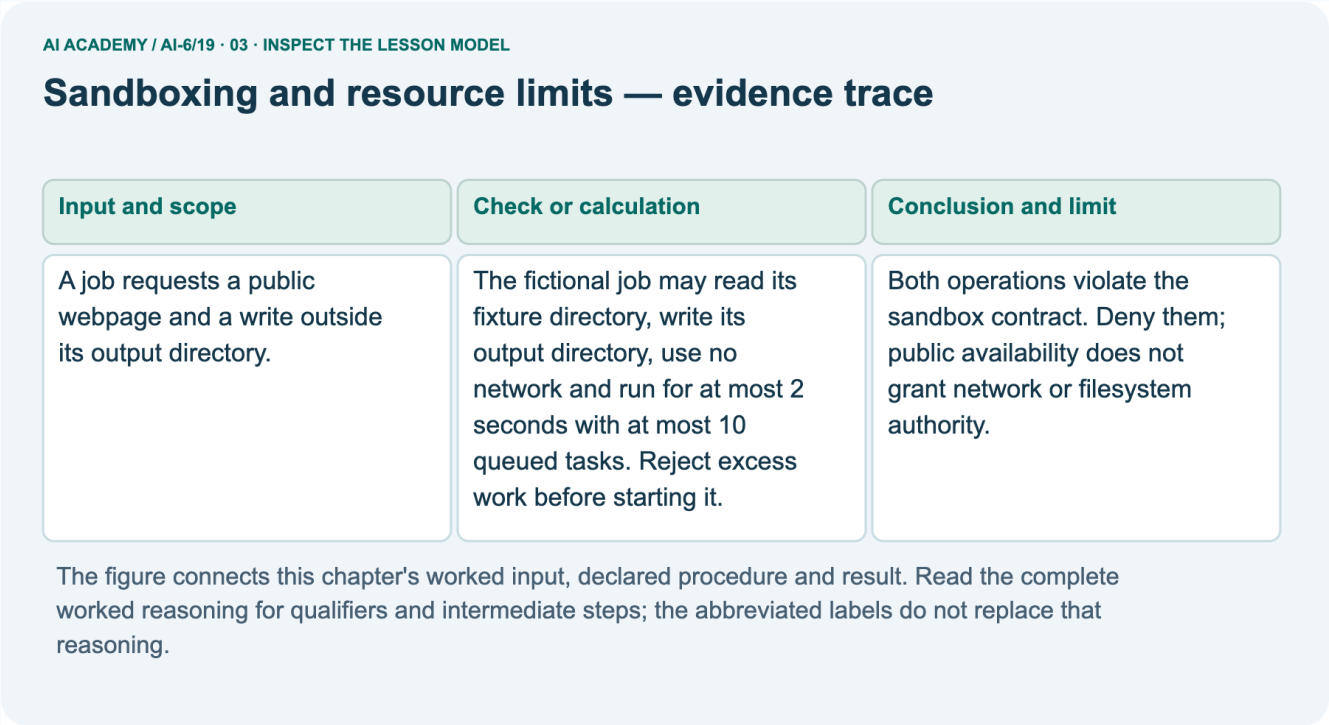
Follow the labelled picture with the matching explanation. Open the vector diagram

Apply isolation: Neither program receives network permission. → Check A: One second and 2 MB fit both resource limits. → Deny B network call: Stop the prohibited network action regardless of its purpose.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Sandboxing and resource limits — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Sandboxing and resource limits — **evidence** trace.

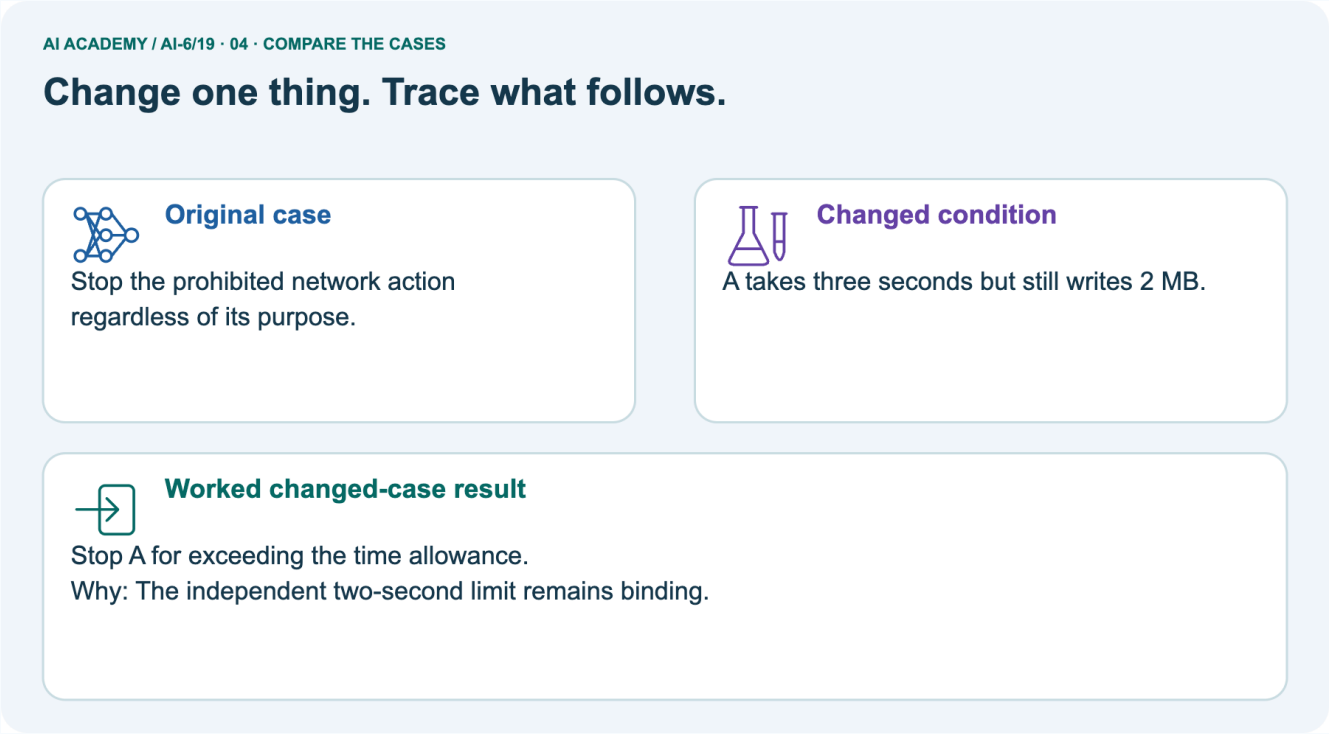
A job requests a public webpage and a write outside its **output** directory. | The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks.

Reject excess work before starting it. | Both operations violate the **sandbox** contract. Deny them; public availability does not grant network or filesystem authority.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

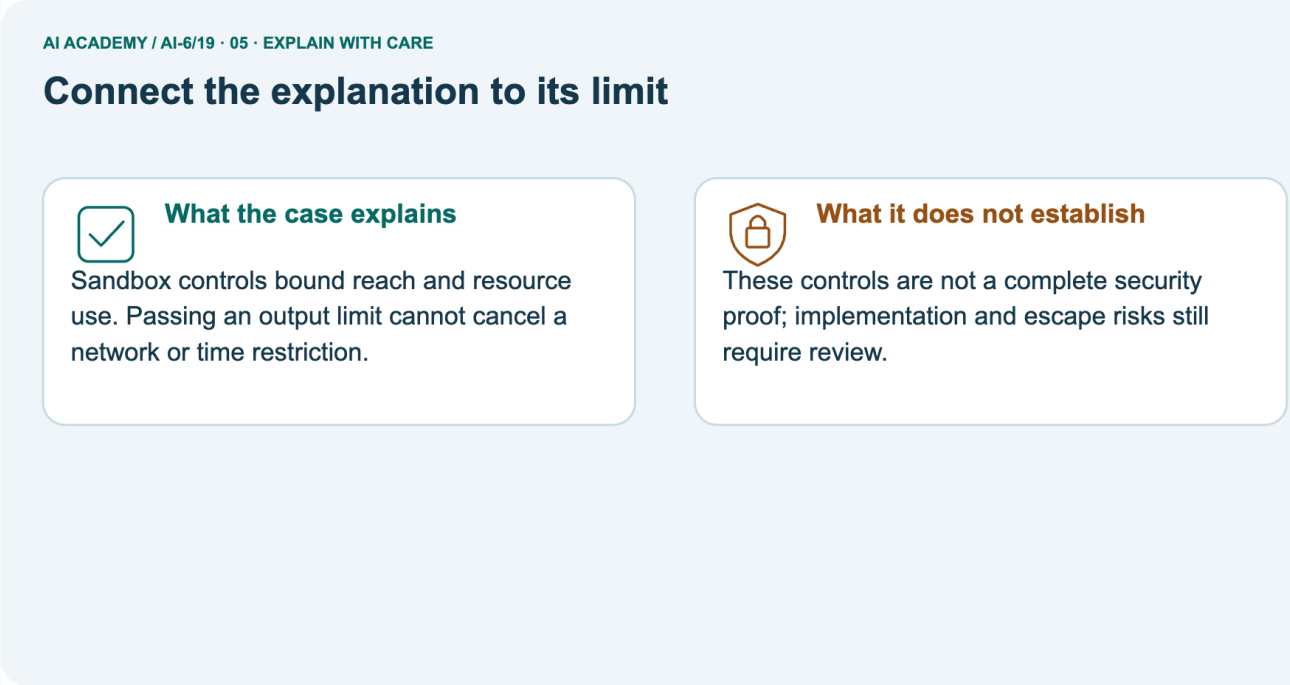
Original case: Stop the prohibited network action regardless of its purpose. Changed condition: A takes three seconds but still writes 2 MB. Expected result: Stop A for exceeding the time allowance.

Reason: The independent two-second limit remains binding.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



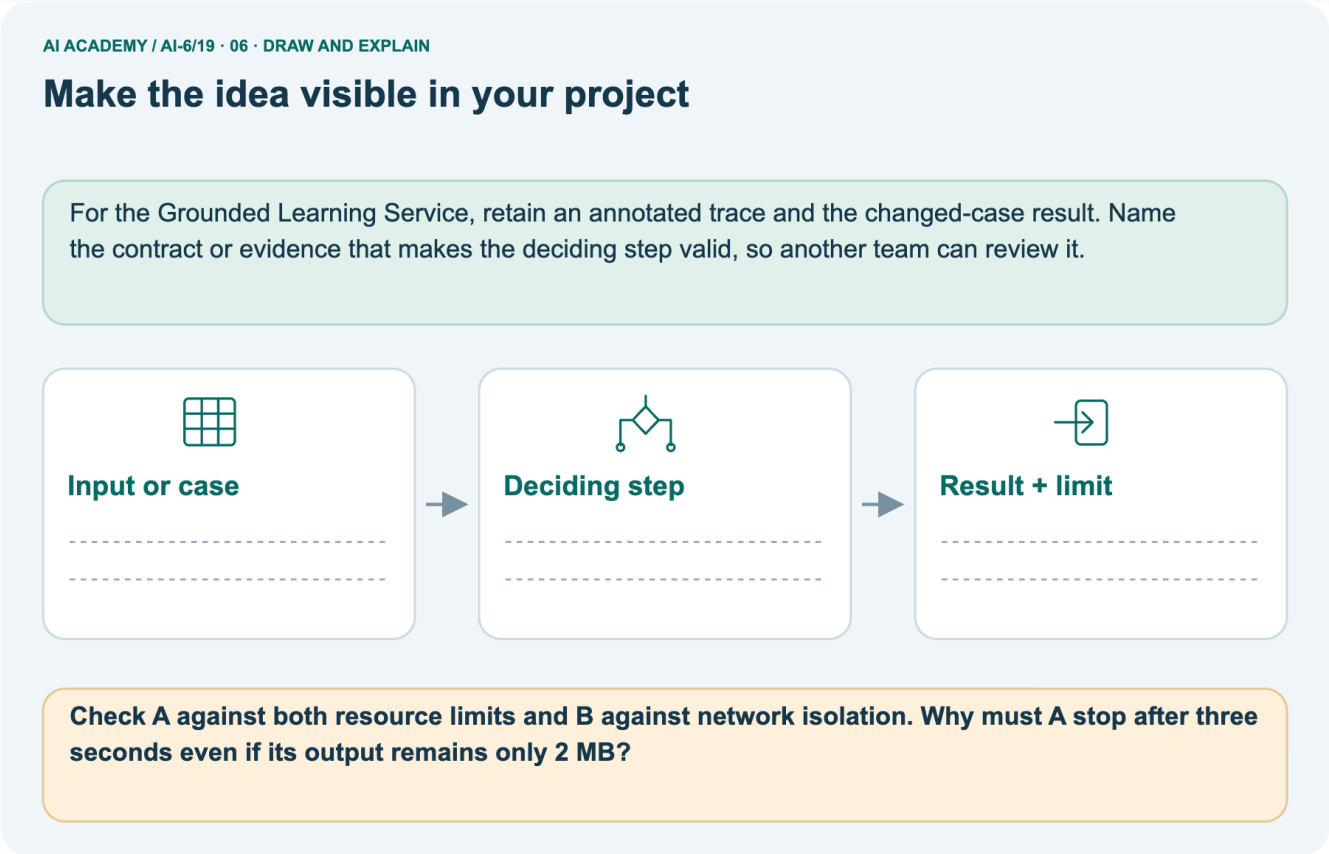
Follow the labelled picture with the matching explanation. Open the vector diagram

Sandbox controls bound reach and resource use. Passing an **output** limit cannot cancel a network or time restriction. Limit: These controls are not a complete security proof; implementation and escape risks still require review.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its **output** remains only 2 MB?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.